

CASE STUDY

DESIGNING A DATA WAREHOUSE SCHEME

FOR E-COMMERCE COMPANIES

1. Background

An e-commerce company, "TokoMaju", wants to build a Data Warehouse to analyze their business performance. Currently, sales transaction data is only stored in the operational system and is difficult to process in the form of analytical reports. Therefore, companies need a Data Warehouse system to support the analysis of sales trends, customer behavior, and product performance.

2. Purpose of Data Warehouse

- Analyze sales performance by product category, region, and time period.
- Understand customer buying patterns, including product preferences and transaction frequency.
- Optimize stock management by understanding product demand patterns.

3. Data Source

The data to be entered into the Data Warehouse comes from various operational systems of the company:

1. Sales Transaction Data is data about every transaction that occurs, including the products purchased, the amount, price, and the payment method.
2. Product Data is the detailed data of products sold, including categories, prices, and brands.
3. Customer Data is data that contains customer information such as location, age, and frequently used payment methods.
4. Time Data is data that contains details of the date of the transaction, such as the day, month, year, and promotion period.

4. Identify Facts and Dimensions

Table 1 Table Identification

Data Type	Category	Information
Sales Facts	Fact	Store analyzeable transaction data (number sold, total revenue, discounts)
Product Dimensions	Dimension	Contains details about the product such as name, category, and price

Data Type	Category	Information
Customer Dimension	Dimension	Contains customer information such as ID, location, and segmentation
Time Dimension	Dimension	Contains transaction time information (date, month, year)
Payment Method Dimensions	Dimension	Contains the payment method information used by the customer

5. Data Warehouse Schema Design

Star Schema Planning

Based on the data that has been identified, the Star Schema scheme can be designed as follows:

Sales Facts (*Fact Table*)

- ID_Penjualan (*Primary Key*)
- ID_Produk (*Foreign Key to Dim_Produk*)
- ID_Pelanggan (*Foreign Key to Dim_Pelanggan*)
- ID_Waktu (*Foreign Key to Dim_Waktu*)
- ID_Pembayaran (*Foreign Key to Dim_Pembayaran*)
- Jumlah_Terjual
- Total_Harga
- Discount

Product Dimension (*Dimension Table*)

- ID_Produk (*Primary Key*)
- Nama_Produk
- Category
- Brand
- Price

Customer Dimension (*Dimension Table*)

- ID_Pelanggan (*Primary Key*)
- Nama_Pelanggan
- Age
- Jenis_Kelamin
- Location

Dimension Table

- ID_Waktu (*Primary Key*)
- Date
- Moon
- Year
- Hari_Dalam_Minggu

Payment Method Dimension (*Dimension Table*)

- ID_Pembayaran (*Primary Key*)
- Jenis_Pembayaran (*Credit Card, Bank Transfer, e-Wallet, COD*)

Task 1

1. Create the following table in the database toko_maju

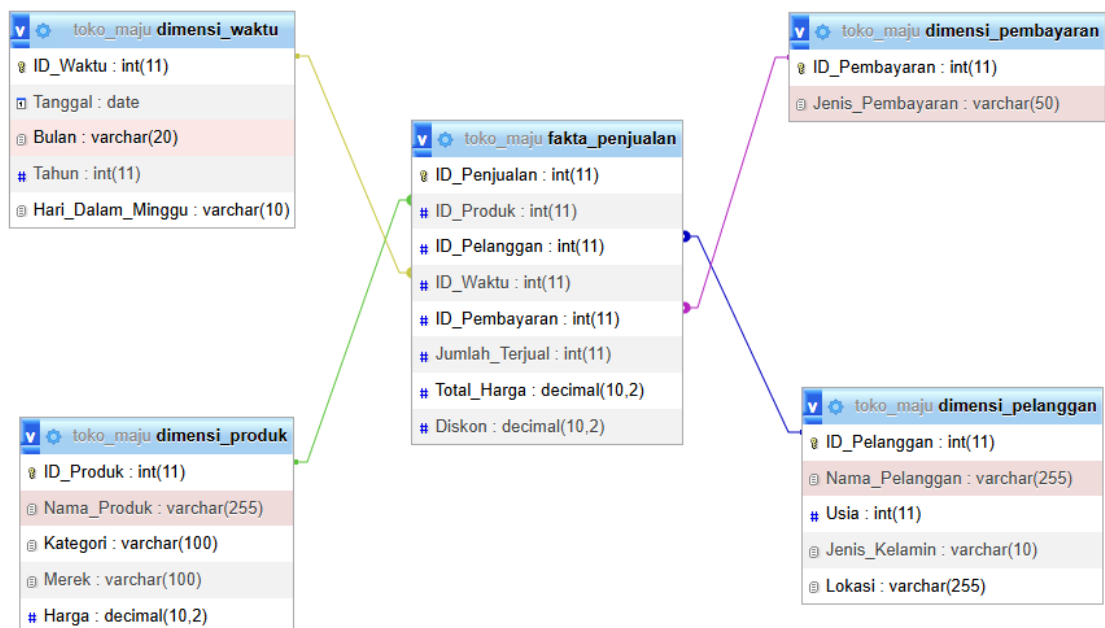


Figure 1. DB Design

You can create it manually directly in the MySQL DBMS or with the following SQL code, **do not** check *Enable foreign key checks* and then click **GO**:

```
CREATE TABLE Fakta_Penjualan (  
    ID_Penjualan INT PRIMARY KEY,  
    ID_Produk INT,  
    ID_Pelanggan INT,  
    ID_Waktu INT,  
    ID_Pembayaran INT,  
    Jumlah_Terjual INT,  
    Total_Harga DECIMAL(10.2),  
    DECIMAL Discount(10.2),  
    FOREIGN KEY (ID_Produk) REFERENCES  
Dimensi_Produk(ID_Produk),  
    FOREIGN KEY (ID_Pelanggan) REFERENCES  
Dimensi_Pelanggan(ID_Pelanggan),  
    FOREIGN KEY (ID_Waktu) REFERENCES  
Dimensi_Waktu(ID_Waktu),
```

```

        FOREIGN KEY (ID_Pembayaran) REFERENCES
Dimensi_Pembayaran(ID_Pembayaran)
    );

CREATE TABLE Dimensi_Produk (
    ID_Produk INT PRIMARY KEY,
    Nama_Produk VARCHAR(255),
    Category VARCHAR(100),
    Brand VARCHAR(100),
    DECIMAL Price(10.2)
);

CREATE TABLE Dimensi_Pelanggan (
    ID_Pelanggan INT PRIMARY KEY,
    Nama_Pelanggan VARCHAR(255),
    INT Age,
    Jenis_Kelamin VARCHAR(10),
    VARCHAR Locations(255)
);

CREATE TABLE Dimensi_Waktu (
    ID_Waktu INT PRIMARY KEY,
    Date DATE,
    On the Moon(20),
    INT Year,
    Hari_Dalam_Minggu VARCHAR(10)
);

CREATE TABLE Dimensi_Pembayaran (
    ID_Pembayaran INT PRIMARY KEY,
    Jenis_Pembayaran VARCHAR(50)
);

```

2. Enter a database on each table

There are 10 following data, please enter the datawarehouse dimension table:

```

INSERT INTO Dimensi_Produk (ID_Produk, Nama_Produk,
Category, Brand, Price) VALUES
(1, 'Laptop A', 'Electronic', 'BrandX', 10000000),
(2, 'Smartphone B', 'Electronic', 'BrandY', 5000000),
(3, 'C Fan', 'Home Appliances', 'BrandZ', 750000),
(4, 'TV LED D', 'Elektronik', 'BrandX', 4000000),
(5, 'E Washing Machine', 'Household Appliances', 'BrandW',
3500000);
(6, 'Headphone F', 'Electronic', 'BrandY', 1500000),
(7, 'Gas Stove G', 'Home Appliances', 'BrandZ',
1200000),
(8, 'Smartwatch H', 'Electronic', 'BrandX', 2000000),
(9, 'Blender I', 'Home Appliances', 'BrandW', 850000),

```

```
(10, 'DSLR Camera J', 'Electronic', 'BrandV', 7500000);
```

Enter the following data in your table:

1. Dimensi_pembayaran

ID_Pembayaran	Jenis_Pembayaran
1	Kartu Kredit
2	Transfer Bank
3	e-Wallet
4	COD
5	Virtual Account
6	Debit Card
7	PayLater
8	Kartu Debit
9	QRIS
10	Paypal

2. Dimensi_pelanggan

ID_Pelanggan	Nama_Pelanggan	Usia	Jenis_Kelamin	Lokasi
1	Ahmad	30	Laki-laki	Jakarta
2	Siti	25	Perempuan	Bandung
3	Budi	40	Laki-laki	Surabaya
4	Rina	35	Perempuan	Yogyakarta
5	Tono	28	Laki-laki	Semarang
6	Dewi	27	Perempuan	Jakarta
7	Rahmat	34	Laki-laki	Surabaya
8	Indah	29	Perempuan	Bandung
9	Hendra	42	Laki-laki	Medan
10	Sari	31	Perempuan	Yogyakarta

3. Dimensi_produk

ID_Produk	Nama_Produk	Kategori	Merek	Harga
1	Laptop A	Elektronik	BrandX	10000000.00
2	Smartphone B	Elektronik	BrandY	5000000.00
3	Kipas Angin C	Peralatan Rumah Tangga	BrandZ	750000.00
4	TV LED D	Elektronik	BrandX	4000000.00
5	Mesin Cuci E	Peralatan Rumah Tangga	BrandW	3500000.00
6	Headphone F	Elektronik	BrandY	1500000.00
7	Kompor Gas G	Peralatan Rumah Tangga	BrandZ	1200000.00
8	Smartwatch H	Elektronik	BrandX	2000000.00
9	Blender I	Peralatan Rumah Tangga	BrandW	850000.00
10	Kamera DSLR J	Elektronik	BrandV	7500000.00

4. Dimensi_waktu

ID_Waktu	Tanggal	Bulan	Tahun	Hari_Dalam_Minggu
1	2024-01-15	Januari	2024	Senin
2	2024-02-10	Februari	2024	Sabtu
3	2024-03-05	Maret	2024	Selasa
4	2024-04-20	April	2024	Jumat
5	2024-05-12	Mei	2024	Minggu
6	2024-06-15	Juni	2024	Sabtu
7	2024-07-20	Juli	2024	Minggu
8	2024-08-08	Agustus	2024	Kamis
9	2024-09-18	September	2024	Rabu
10	2024-10-25	Oktober	2024	Jumat

Filling in the data into the fact table there are **15 transactions that refer to the previous dimension table**

```
INSERT INTO Fakta_Penjualan (ID_Penjualan, ID_Produk,
ID_Pelanggan, ID_Waktu, ID_Pembayaran, Jumlah_Terjual,
Total_Harga, Discount) VALUES
(1, 1, 1, 2, 2, 20000000, 500000),
(2, 2, 2, 2, 3, 1, 5000000, 250000),
(3, 3, 3, 3, 4, 3, 2250000, 150000),
(4, 4, 4, 4, 1, 1, 4000000, 200000),
```

```
(5, 5, 5, 5, 5, 2, 7000000, 350000);
(6, 6, 6, 6, 6, 3, 4500000, 300000),
(7, 7, 7, 7, 7, 2, 2400000, 120000),
(8, 8, 8, 8, 8, 1, 2000000, 150000),
(9, 9, 9, 9, 9, 4, 3400000, 200000),
(10, 10, 10, 10, 10, 1, 7500000, 500000),
(11, 1, 2, 3, 1, 2, 20000000, 750000),
(12, 2, 4, 5, 2, 1, 5000000, 200000),
(13, 3, 6, 7, 3, 5, 3750000, 300000),
(14, 4, 8, 9, 4, 1, 4000000, 250000),
(15, 5, 10, 1, 5, 3, 10500000, 500000);
```

3. Data analysis is complemented by SQL code/query and its table output.

1. Total Sales per Product Category, example SQL code:

```
SELECT dp. Category, SUM(fp. Total_Harga) AS
Total_Pendapatan
FROM Fakta_Penjualan fp
JOIN Dimensi_Produk dp ON fp.ID_Produk =
dp.ID_Produk
GROUP BY dp. Category;
```

Explanation per Section:

<i>Query</i>	<i>explanation</i>
SELECT dp. Category, SUM(fp. Total_Harga) AS Total_Pendapatan	• Dp. Category → Retrieves the product category from the Dimensi_Produk (dp) table. • SUM(fp. Total_Harga) US Total_Pendapatan → Sum the total price of each transaction based on product category, then the result is given an alias "Total_Pendapatan" for easier reading.
FROM Fakta_Penjualan fp	• This table stores transaction information, including products purchased, customers who purchased, date of purchase, payment method, amount sold, and total price. • fp means fakta_penjualan
JOIN Dimensi_Produk dp ON fp.ID_Produk = dp.ID_Produk	• This section connects Fakta_Penjualan tables with Dimensi_Produk based on ID_Produk columns. • fp.ID_Produk = dp.ID Produk → Links each

	transaction in the Fakta_Penjualan with more detailed product information in Dimensi_Produk .
GROUP BY dp. Category	• This section groups results by product category. • All transactions that have the same product category will be grouped into a single row in the query results. • Then, the Total_Harga value of all transactions in that category will be summed up. • If there are multiple transactions for the Electronics category, the system will add up all Total_Harga values associated with that category. • Likewise for the Home Appliances category and other categories if any.

2. Number of Transactions per Payment Method, example query:

```
SELECT dp. Jenis_Pembayaran,
COUNT(fp.ID_Penjualan) AS Jumlah_Transaksi
FROM Fakta_Penjualan fp
JOIN Dimensi_Pembayaran dp ON fp.ID_Pembayaran =
dp.ID_Pembayaran
GROUP BY dp. Jenis_Pembayaran;
```

3. Monthly Sales Trends in 2024?
4. Total Sales per Month?
5. Product Categories with the Most Sales?
6. Most Frequently Used Payment Methods?
7. Customers with the Highest Total Purchases?

4. Give a Conclusion from this practicum